

Information Integration in Computerized Distribution System Planning

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Abstract— Electric Distribution Planning involves a great deal of information, residing in different systems. Information sharing among these systems is essential in improving the efficiency and quality of distribution system planning. This paper presents an integrated solution for computerized distribution planning in a Geographic Information System (GIS) context, a synergy that magnifies the data accessibility between a load forecasting and feeder planning tools, sealing the traditional gap between long-term and short-term distribution system planning. A stochastic cell-based load forecasting algorithm is first developed, followed by an optimal load allocation module, NODESIM, which spatially relates the load growth to the vector-based circuit topology from feeder planning tools. NODESIM enables the multi-year distribution system studies in a GIS context, to best assist utility planners in deciding where and when the customers will grow and how to expand the system facilities to meet the demand growth.

I. INTRODUCTION

In the competitive environment, the utility needs to minimize the overall costs of its delivery system. It can achieve this goal by minimizing the capacity requirements and trading off capital expenditures against operating costs. To do so involves the coordinated implementation of three functions [1]:

1. Spatial load forecasting
2. Long range distribution planning
3. Short range distribution planning

A. Spatial load forecasting

Spatial load forecasting projects not only the magnitude but also the location of future load growth. The forecast establishes the loading requirements that the long and short range planning must meet. The spatial load forecasting is best done on the geographic information system (GIS) platform, analyzing and forecasting customer demands in each geographic small area or cell. This type of methods avoid the pitfalls of forecasting by substation of feeder service area:

- no historical data -- growth often occurs far from existing equipment,
- land use saturation -- trending by substation and feeder loads often fails to recognize land availability that limits load growth, and
- load transfer -- transfer of loads from one substation or feeder to another is difficult to capture in the trending substation or feeder loads.

B. Long Range Planning

The long range planning does not lead to immediate decisions on equipment and site acquisition. Rather, its purpose is to set the direction for the short range planning decisions—in the long run, where substation and capacities should be added. The location aspect of the long range planning is the overriding factor. If the equipment size is off, the utility can often adjust the purchase at the time. If the timing is slightly off, the utility can advance or delay the construction by a year or two. On the other hand, if the location is off, the utility may have acquired substation sites and new right-of-ways in wrong locations. The result would be longer subtransmission lines and feeders, leading to much higher costs. Hence, long range planning should be done in the GIS context, the same as spatial load forecasting.

C. Short Range Planning

In the short range planning function, an engineer must decide on equipment size and settings, and the precise location of the equipment, both geographically and electrically. Hence the function requires the precise modeling of the distribution system, by nodes and lines.

D. Coordination of Long and Short Range Planning

This coordination is necessary to make sure that (1) short range planning decisions follow the direction set by long range planning, (2) long range planning adequately recognizes the constraints of the existing system and near-term needs, and (3) both long and short term plans meet the forecasted demand. As discussed above, the three functions are often carried out in different data models—a GIS based spatial model for forecasting and long range planning, and

This paper was presented at the 1996 IEEE Transmission and Distribution Conference held in Los Angeles, California, September 15–20, 1996.

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an equipment line-node model for short range planning. So, coordination of the three functions requires an integration of the different data models. This paper proposes a method of such an integration.

The differences in the planning models used in long and short range planning environments are identified. Then, a stochastic landuse simulation forecasting algorithm is introduced. To effectively utilize the forecasting results, a database integration interface utilizing the latest client-server implementation of Relational Database Management Systems (RDBMS) is developed. The authors also propose a heuristic based algorithm (A* like) [2-3] for spatial load aggregation-allocation so the long term load forecasting results can be referenced in the short term feeder planning tool in an automated manner.

II DATA MODEL AND NOTATIONS

Many utilities now use small area analysis for long range load forecasting. The results are often kept in a set of rasters (**Definition of Raster**: A format for storing, processing, and displaying graphic data in which graphic images are stored as values for uniform grid cells or pixels [4]). Feeder planning tools, on the other hand, tend to reside in a vector based drawing environment because the feeders/points can be well represented with a group of arcs and nodes [4]. Figure 1 illustrates the different planning models.

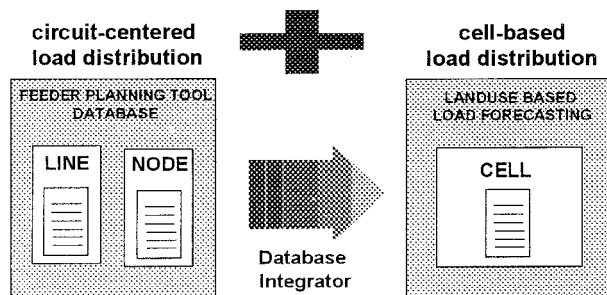


Figure 1 Different Planning Models

The notations used in this paper are:

V: vector set **R**: raster set **R**: real number set
 $\mathcal{R}^{(s,A)}$: $V \rightarrow R \rightarrow R^{n \times m}$; rasterization performed with size s square and for area **A**.

III. STOCHASTIC SPATIAL LOAD FORECASTING

Small-area analysis methods have been widely adopted as the primary instrument to gather long range load forecasting results [5-6]. As new technologies evolve, small area analysis methods are no longer a time-consuming or complex process for distribution planners; many of them

have been computerized in an automated environment. The progressive development in Geographic Information Systems (GIS) has dramatically improved the traditional cell-based spatial analysis, automating most time-consuming manipulations in an efficient manner. At the same time, accuracy of small-area analysis is elevated because higher resolution cell-based processing is made possible in modern computing environment. The first step in small area analyses is to rasterize vector maps with respect to certain attributes: the **Theme**. Theme based rasterization is a process of relating the associations between rasters and vectors, which is defined as:

$$\mathcal{R}(MAP(theme)) = \bigcup_{i,j}^{theme} C(i,j) = \{C(i,j)\}_{theme} \quad (1)$$

$$\xrightarrow{theme} \{theme(i,j)\} = THEME$$

For example, the landuse map $MAP(landuse)$ is originally a vector with a set of polygons; each polygon has a landuse attribute. After the rasterization process, a set of uniform cells $C(i,j)$ are created as a raster. The raster is mapped to a matrix ($n \times m$): **LANDUSE**, using the landuse attribute as the cell value. Let $lc(l, t)$ be the saturated coincident load curve for the landuse class l at time t . A load map at $t = T$ (assuming the system peaks at T) is simulated by the following mapping function l :

$$l: R^{n \times m} \rightarrow R^{n \times m} \quad (2)$$

$$l(landuse(i,j)) \xrightarrow{lc(l,T)} load(i,j)$$

l is a linear mapping if $lc(l, t)$ exists and

$$lc(l, t) = \lim_{N \rightarrow \infty} \frac{LC(l, t)}{N}, \quad (3)$$

where N = number of landuse l customers sampled and $LC(l, t)$ is the metered coincident load

The electric demand growth/change from the long-term perspective is dictated by two factors: new customers, and changes in usage. To simulate the load growth, the $lc(l, t)$ is trended to represent the usage changes; however, the landuse changes are rather complex to simulate. The land-use simulation in load forecasting context is to mimic how the landuse will change given the set of geographic factors (highways, roadways, surrounding environment, and others) and regulatory constraints (zoning). The urban development model can be very different from one region to the other. Normally, a combination of linear (proximity, f_l), nodal (convolution, f_n), and zonal (surrounding environment, f_z) models is applied to emulate the preference, p , of how a particular landuse develops. The landuse simulation function, s , is based on the p , f , and other stochastic nature of landuse development. These functions are defined:

$s: \mathbf{R}^{n \times m} \rightarrow \mathbf{R}^{n \times m}$, landuse simulation function
 $f: \mathbf{R}^{n \times m} \rightarrow \mathbf{R}^{n \times m}$, factor creation function ($f_1/f_2/f_3$)
 $p: \mathbf{R}^{n \times m} \rightarrow \mathbf{R}^{n \times m}$, preference creation function

$$\text{LANDUSE}^i = s(\text{LANDUSE}^{i-1}), \text{ for the } i\text{th iteration} \quad (4)$$

$$\mathbf{F}_j^i = f(\text{THEME}^i), \quad (5)$$

factor j ($1 \leq j \leq J$) for iteration i

$$\mathbf{P}_l^i = \sum_{j=1}^J w_j \times \mathbf{F}_j^i, \text{ preference for landuse } l \text{ and} \quad (6)$$

iteration i , and w_j is the weighting scalar for factor j

A new stochastic landuse simulation algorithm is derived from landuse-based load simulation by adding the Markov process model to handle the stochastic nature of urban development. If the total number of different landuse classes is L , a stochastic (one-step transition) matrix $\mathbf{P}$ can be established with transition probabilities $p(i,j)$'s [7].

$$p^n(i,j) = p\{X(t_n) = j | X(t_{n-1}) = i\} \text{ represents} \quad (7)$$

the conditional probability of the landuse being j at time t_n , given it was i at time t_{n-1}

$$p^n(i,j) \geq 0; 1 \leq i \leq L \text{ and } 1 \leq j \leq L$$

$$\sum_{j=1}^L p^n(i,j) = 1$$

The absolute (state) probabilities of iteration i are defined as:

$$\mathbf{P}_S^i = \mathbf{P}_S^{i-1} \mathbf{P}^i, \text{ where } \bar{\mathbf{P}}_S^i = [p_S^i(1) \ p_S^i(2) \ \dots \ p_S^i(L)] \quad (8)$$

Stochastic landuse based load forecasting algorithm is summarized as follows:

A. Deciding Landuse Allocation Order

If there is no urban redevelopment for iteration i , only the vacant landuse ($l = L$) can be changed to other landuses. The stochastic matrix, $\mathbf{P}^i$, is defined as:

$$\mathbf{P}^i = \begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \vdots & \vdots \\ 0 & 0 & 1 & 0 & \vdots \\ \vdots & \dots & 0 & 1 & 0 \\ p^i(L,1) & p^i(L,2) & \dots & \dots & p^i(L,L) \end{bmatrix}, \quad (9)$$

$$\text{where } \sum_{j=1}^L p^i(L,j) = 1 \text{ and } p^i(L,j) \geq 0$$

If urban redevelopment is taken into account (allowing landuse conversion between non-vacant landuses), $\mathbf{P}^i$ can be defined as:

$$\mathbf{P}^i = \begin{bmatrix} p^i(1,1) & \dots & p^i(1,L) \\ \vdots & \ddots & \vdots \\ p^i(L,1) & \dots & p^i(L,L) \end{bmatrix}, \text{ where } \sum_{j=1}^L p^i(n,j) = 1, \quad (10)$$

$p^i(n,j) \geq 0$ for $\forall 1 \leq n \leq L$, for $1 \leq n \leq L-1$, $p^i(n,n) \approx 1.0$ for $1 \leq n \leq L-1$, $1 \leq j \leq L-1$, and $n \neq j$, $p^i(n,j) \approx 0.0$

The stochastic matrix $\mathbf{P}^i$ in this case allows landuse classes other than vacant land to convert to others although the probabilities are low. A landuse distribution vector is defined for the $(i-1)$ iteration as:

$$\bar{\mathbf{L}}^{i-1} = [\bar{l}_1^{i-1} \ \bar{l}_2^{i-1} \ \dots \ \bar{l}_L^{i-1}], \text{ where } \bar{l}_1^{i-1} \text{ represents the percentage distribution of landuse } 1, \text{ where } \bar{\mathbf{L}}^{i-1} \in \mathbf{R}^L \text{ and } \sum_{j=1}^L \bar{l}_j^{i-1} = 1$$

If $\bar{\mathbf{L}}^i$ is known (global landuse changes are trended)

$\bar{\mathbf{L}}^{i-1} \mathbf{P}^i = \bar{\mathbf{L}}^i$. Because $\mathbf{P}^i$ is linerly independent ($\because p^i(L,j) \geq 0$),

$\{p^i(L,j)\}$ will have a unique non-trivial solution.

If $\{p^i(L,j)\}$ is defined, $\bar{\mathbf{L}}^i$ is calculated (probablistic landuse forecast).

B. Simulating Landuse Changes

sort: $\mathbf{R}^n \rightarrow \mathbf{R}^n$ is a function to sort all the elements in a vector so they are placed in the descendent order. The landuse change simulation is described in Figure 2.

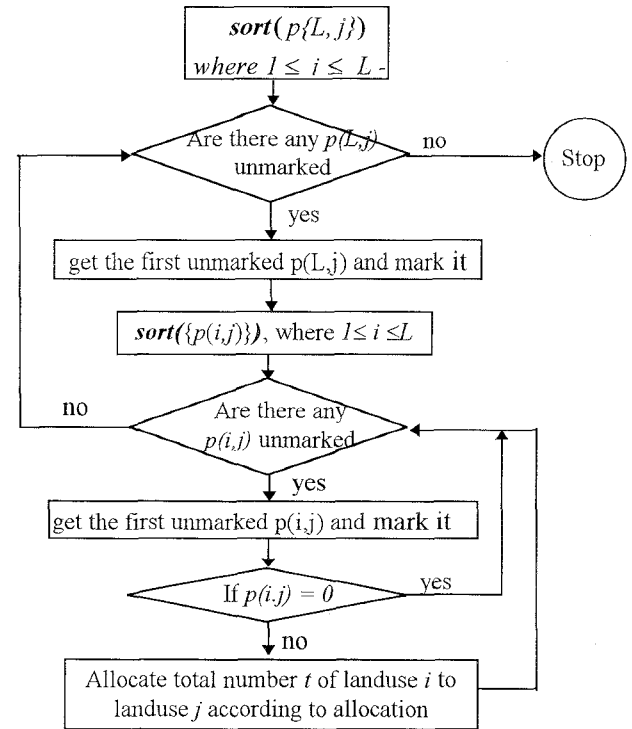


Figure 2 The Landuse Simulation Algorithm

Note: $t = m \times n \times \left(p^i(i, j) \times p_s^{i-1}(i) \right)$

C. Allocating Strategies

TEST: $\mathbf{R}^{n \times m} \rightarrow \mathbf{R}^{n \times m}$ is a function to create a logic grid map and defined as:

$$\text{TEST}(\text{MAP}, k) = \begin{cases} \text{test}(i, j) = 1, & \text{if map}(i, j) \text{ satisfies } k \\ \text{test}(i, j) = 0, & \text{if map}(i, j) \text{ does not satisfy } k \end{cases}$$

k is a logical expression.

For $\text{landuse} = l$, a SCORE is formed by combining the
 $\text{SCORE} = \text{TEST}(\text{LANDUSE}^i, \text{landuse} = j) \times \text{TEST}(\text{ZONING}^i, \text{zoning} = l) \times P_l^i$ (11)

Allocate total t of landuse j to landuse l based on the $\text{score}(i, j)$.

III. NODESIM ALGORITHM

NODESIM is a module that relates small-area load forecasting results to the circuit topology, so the distribution planner can assess the short-term feeder expansion plans with the long-range forecasting results. Functionally, NODESIM contains the following major processes:

- (1) database integrator
- (2) topology creation and referencing
- (3) optimal load aggregation (cell load to bus load) based on effort maps
- (4) strategic load distribution (to different load buses in one cell)
- (5) bus loading analyses, including calculation and visualization of site (substations, load buses) loading

A. Establish associations between node and load

Electric attributes are centered on load. The SQL techniques are used to retrieve electric information from the tables of feeder planning tool. The hierarchy of ALLOCATED LOAD, LOAD, and NODE is describe in Figure 3 (based on OMT: Object Modeling Techniques [8]):

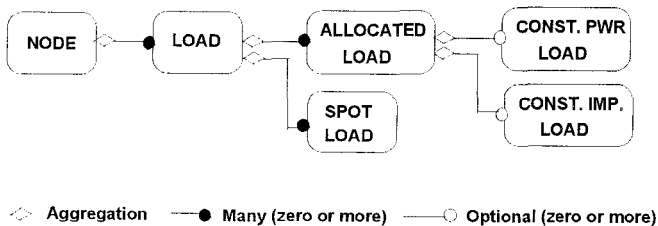


Figure 3 Load Model

In plain English, the model indicates that a NODE would have one or more than one LOAD associated with it; LOAD could be composed of one or more than one ALLOCATED LOAD/SPOT LOAD and each ALLOCATED LOAD consists of zero or one CONST. PWR LOAD/COSNST. IMP. LOAD.

B. Identify load bus cell

The circuit topology is established through the database connection. Geographically, the circuit is referenced to the study area as depicted in Figure 4. In this way, a **BUS** grid map is produced and defined as:

$$\text{BUS} = \begin{cases} \text{bus}(i, j) = 1 & \text{if there are one or more than one load points} \\ \text{bus}(i, j) = 0 & \text{if there is no load point} \end{cases}$$

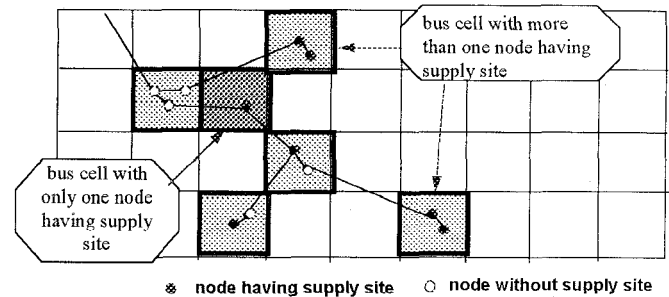


Figure 4 Identification of Bus Cell

C. Optimize load cell (load growth) to bus cell assignment

To relate the load forecasting results to the existing circuit, a load growth map is first created. The load-to-bus cell assignment identifies where the load growth takes place in the existing circuit. A **EFFORT** map is required to provide cost information with respect to the geographic distribution of the customers. Figure 5 depicts the underlying routing task. The routing task is synthesized using A* search algorithm [2-3] to find the most reasonable load point for the load growth to occur (please refer to Appendix A for proof of optimality). Figure 6 outlines the optimal load growth assignment algorithm.

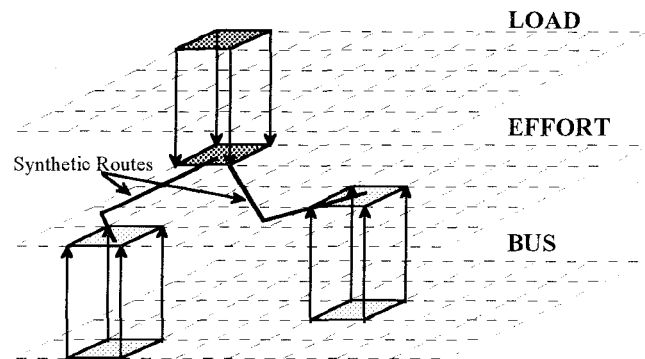


Figure 5 Synthetic Routing on Layers of Grid Maps

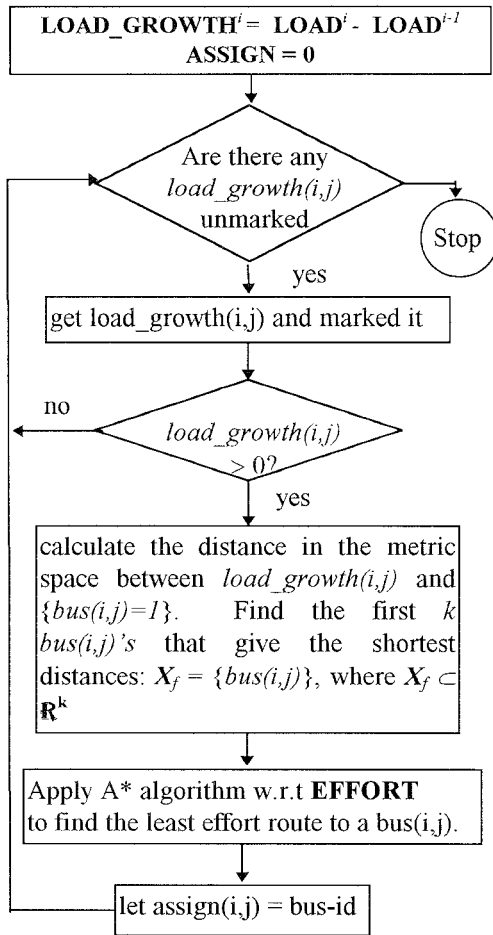


Figure 6 Load-to-bus Cell Assignment Algorithm

D. Aggregate load growth

For each bus cell, the allocated load is updated with the load growth that is assigned to the bus.

$$bus_load^i(b) = bus_load^{i-1}(b) + \sum_{\{assign(i,j)=b\}} load_growth(i,j)$$

F. Strategic load allocation

If the bus cell contains more than one load points, there are several ways to handle the load allocation according to the study propose:

1) user-specified scalars

Each load point from feeder planning tool can be specified with certain scalar to resolve the load allocation conflicts. For example, all the load points on the lower voltage lines will be less preferable to allocate the new load growth if the whole system is going through a voltage upgrade.

2) even distribution

Even distribution provide the user with the easiest way to allocate the load growth when multiple load points existing in one bus cell. It is useful for systems with plenty of unused capacities.

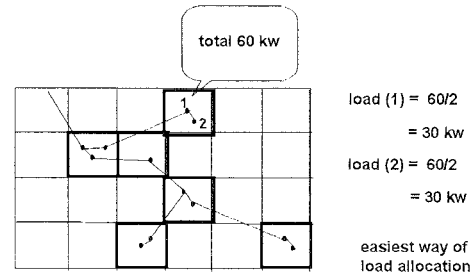


Figure 7 Even Distribution of Multiple Load Points

3) proportional to connected kva

The available connected KVA is often used for load allocation weighting quantity. It follows the natural system expansion trend. In most cases, the result should be good for longer range of planning.

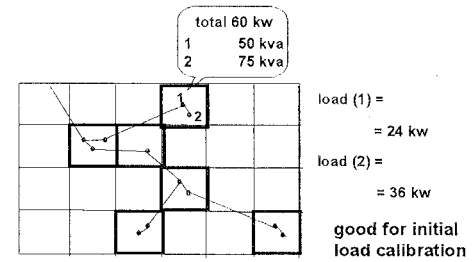


Figure 8 Connected KVA based Load Allocation

4) reciprocal to capacity utilization

The current utilization of load points is a good measurement for how the expansion plan is going to develop. This method is good for short range feeder expansion planning when many of the load points are operated at full capacities.

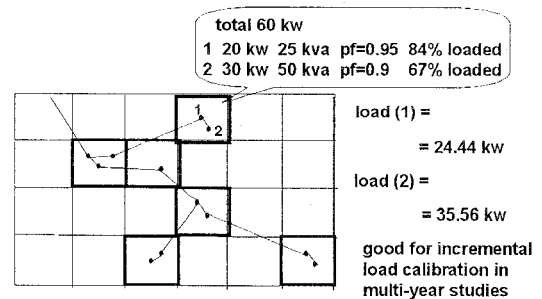


Figure 9 Utilization Based Load Allocation

E. Send results to the present circuit database

The load growth results are sent back to the feeder planning tool for further analyses.

V. IMPLEMENTATION AND RESULTS

The system is implemented in a client-server database environment so the data transaction is very flexible. Figure 10 shows the system implementation structure. A commercial GIS system as the spatial manipulation host environment. Most the algorithms are implemented in C programming language. Figure 11 shows the identified bus cell after a study area is referenced by a circuit topology. Figure 12 displays the results of the optimal load-to-bus assignment (grey scale indicating different bus service areas).

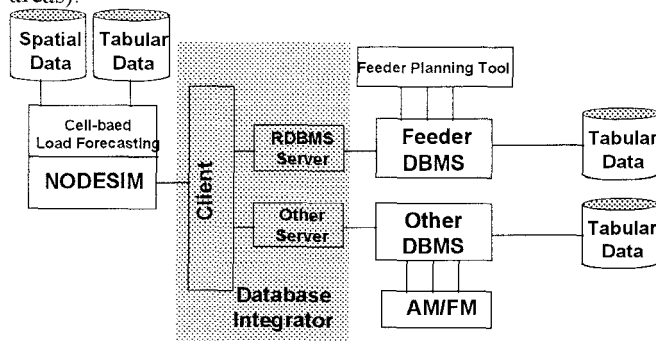


Figure 10 Client-Server Implementation of NODESIM

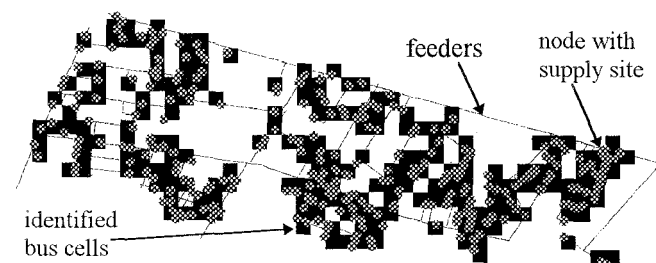


Figure 11 Bus Cell Identification

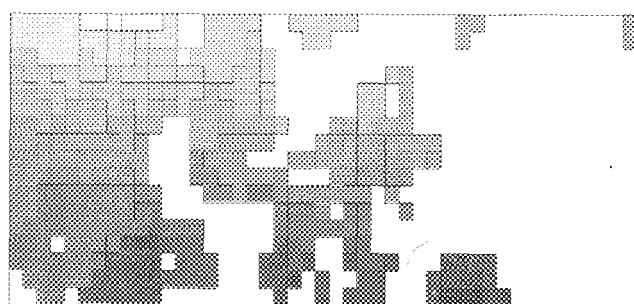


Figure 12 Optimal Load-to-bus Assignment

V. CONCLUSIONS

The authors have presented a stochastic load forecasting algorithm and a method to integrate the spatial GIS based data model used in load forecasting and long range distribution planning with the equipment line-node data

model used in short range planning. This method is efficient and has been successfully implemented in utilities worldwide. The authors plan to publish some of these application case studies in future papers.

VI. REFERENCES

- [1] M. V. Engel et al., *Distribution Planning, Tutorial*, IEEE Summer Meeting 1993, Vancouver, B. C., Canada, July 1993.
- [2] K. Passino and P. J. Antsaklis, "A Metric Space Approach to the Specification of the Heuristic Function for the A* Algorithm, IEEE Transactions on Systems, Man, and Cybernetics, Vol. 24, No.1, Jan., 1994.
- [3] E.-C. Yeh, Z. Sumic, and S. S. Venkata, "An Intelligent Primary Router for Underground Residential Distribution Design," Proceedings of Fifth Intelligent Systems Application to Power Systems '94, France, Sept. 1994 (Accepted for Publication by Engineering Intelligent Systems Journal, 1996).
- [4] J. C. Antenucci et al, *Geographic Information Systems-A Guide to the Technology*, Van Nostrand Reinhold, New York, 1991.
- [5] H. Lee Willis and Gary B. Rackliffe, *Introduction to Integrated Resource T&D Planning*, ABB Systems Control, Automated Distribution Division, Cary, NC, 1994.
- [6] Electric Power Research Institute, *Research Into Load Forecasting and Distribution Planning*, EPRI EL-1198, Volume 1, August 1980.
- [7] T. Gonen, *Electric Power Distribution Engineering*, McGraw-Hill, 1986.
- [8] M. Keller and K. Shumate, *Software Specification and Design*, John Wiley & Sons, Inc., New York, 1992.
- [9] J. Rumbaugh et al., *Object-Oriented Modeling and Design*, Prentice Hall, Englewood Cliffs, New Jersey, 1991.

VII. BIOGRAPHIES

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